

Predictive Maintenance Pilot - Vibration & Acoustic Analytics

Tampines Data Centre 1 · Data Centre · Singapore

Document ID	DC-03
Category	Predictive Maintenance
Classification	Technical
Date	2025-10
Prepared by	Data Centre Engineering & Analytics

1. Pilot scope

Time-based maintenance was not preventing unplanned failures of rotating equipment at TDC1: four failures of pumps and CRAH fans in 2024 each needed an emergency call-out. A nine-month pilot instrumented 116 critical assets with wireless vibration and acoustic sensors.

Asset group	Units instrumented
Chillers (compressors and motors)	6
Chilled-water and condenser-water pumps	24
CRAH fans	86

2. How it works

- Sensors sample every 10 minutes and capture a high-resolution vibration burst once a day.
- An edge gateway computes spectral features, including harmonics of running speed and bearing defect frequencies.
- A machine-learning model learns each asset's normal baseline and flags anomalies with a severity score.
- Every alert creates a condition-based work order in the CMMS, with the spectrum attached for the technician.

3. Results

Fault detected	Asset	Lead time before failure
Bearing wear	Condenser-water pump P-07	6 weeks
Bearing wear	CRAH fan DH1-14	4 weeks
Bearing wear	Chiller CH-02 motor	5 weeks
Shaft misalignment	Chilled-water pump P-03	3 weeks
Shaft misalignment	CRAH fan DH3-02	2 weeks
Belt looseness	CRAH fan DH2-21	2 weeks
Cavitation	Condenser-water pump P-11	3 weeks

Mean time between failures across the instrumented assets rose by 38%. Maintenance cost fell 15% because fixed-interval overhauls were replaced by repairs triggered by condition.

4. False alarms

In the first month 22% of alerts were false alarms. Technicians labelled every alert as confirmed or false, and the labels were used to retune thresholds for each asset; the false-alarm rate fell to 6% by month four.

5. Lessons learned

- Start with the critical rotating equipment whose failure hurts most.
- The technician feedback loop is essential: labelled alerts turn a noisy system into a trusted one.
- Integrate with the CMMS from day one so every alert has an owner.
- Wireless sensors can be retrofitted without shutting equipment down.

6. Cost and scale-up

The installed cost was S\$400-650 per monitored asset plus an analytics subscription. The approach suits any motor-driven equipment with a characteristic vibration signature - pumps, fans, compressors and traction machines - and is being scaled to all rotating plant at TDC1.